Outlier-Robust Centralized and Distributed Variational Bayesian Moving Horizon Estimation

Zihao Jiang, Giorgio Battistelli, Luigi Chisci, *Life Senior Member, IEEE*, Nicola Forti, and Weidong Zhou

***Abstract*—This paper addresses robust adaptive state estimation for a linear dynamical system subject to measurement outliers and Gaussian noises with unknown time-varying covariances. The proposed method relies on a Bernoulli-Gaussian (BG) model of measurement noise as well as on a variational Bayesian-moving horizon estimation (VB-MHE) approach that allows to jointly estimate the state trajectory on a moving window of fixed size along with the noise parameters (outlier probability, process noise and measurement noise covariances). A centralized robust adaptive filter is first derived and then extended to the distributed multi-sensor case by exploiting a distributed variational Bayesian (DVB) approach. It is shown, via simulation experiments, how the performance of both proposed, centralized and distributed, robust adaptive filters is very close to the ideal one achievable by a Kalman filter with perfect outlier detection and full knowledge of noise covariances.**

***Index Terms*—Process noise covariance matrix, moving horizon estimation, Bernoulli-Gaussian model, distributed variational Bayesian, outlier detection.**

I. Introduction

STATE estimation is a task of paramount importance with many relevant engineering applications, e.g., in target tracking, navigation and control systems. Whenever the system model is linear and Gaussian with known noise covariances, state estimation admits an optimal solution which is recursively implemented by the Kalman filter. In many practical situations, however, the *process noise covariance matrix* (PNCM) and/or *measurement noise covariance matrix* (MNCM) are unknown and, for improved performance, could be jointly estimated, along with the system state, from the available data by means of some adaptive Kalman filter [1], [2], [3], [4], [5], [6], [7], [8], [9]. Moreover, sensor measurements may take occasional anomalous values (*outliers*) which, unless their effect is not properly mitigated by some outlier-robust Kalman filter [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], might severely deteriorate estimation performance.

The aim of this work is to develop a robust adaptive state estimation algorithm to cope with the simultaneous presence of outliers and of uncertainty on PNCM/MNCM. In this respect, it is worth to stress that simultaneously counteracting outliers and adapting to the unknown and possibly time-varying noise statistics is a challenging goal as robustness to outliers and adaptation to changes of noise characteristics are actually conflicting objectives. In particular, it can be difficult to distinguish between a sudden outlier and a genuine shift in noise statistics. For instance, a large deviation in measurements might be interpreted as either an outlier or a result of changing process noise covariance, leading to incorrect responses if the wrong assumption is made. This challenge complicates the design of filtering algorithms that must handle both phenomena simultaneously.

In the existing literature, many references addressed either adaptation of noise parameters [1], [2], [3], [4], [5], [6], [7], [8], [9] or robustness to outliers [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], but only a few attempted to simultaneously account for both [21], [22], [23], [24]. In cases where the MNCM is unknown while the PNCM is known, *inverse Gamma*, *inverse Wishart* (IW) and *Wishart* distributions are typically employed as prior for the unknown covariance or information matrix of Gaussian measurement noise [2], [3], [4], thus enabling joint inference for both the state and MNCM. Conversely, handling situations where also the PNCM is unknown is much more challenging because the PNCM is hidden in the predicted density. In this respect, it has been proposed to estimate the one-step prediction error covariance (PECM) instead of the PNCM [4], [5]. In alternative to this approach, it is intuitively feasible to exploit *smoothing*, *sliding window filtering*, or *moving horizon estimation* (MHE) to directly infer process noise statistics [6], [7], [8], [9]. In particular, smoothers and sliding-window filters update the smoothing density with the current measurements to estimate the state trajectory, and then infer the noise statistics by using the estimated state trajectory [6], [7]. In contrast, MHE [8], [9] computes, by means of state augmentation, the joint density of states and noise statistics

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using all measurements within a moving window, resulting in improved estimation accuracy and guaranteed stability properties [8]. However, all these estimation methods typically rely on the *minimum mean square error criterion* and assume Gaussian distributions, thus being sensitive to outliers and unable to ensure filtering accuracy under outlier interference.

Traditional methods for counteracting outliers rely on the χ^2*-test* [10], where outliers are removed if a residual exceeds a given threshold, though this threshold sensitivity can affect estimation accuracy. Methods based on robust estimation criteria, such as the *Huber* [11], *maximum correntropy* [12] and *minimum error entropy* [13] cost functions are also influenced by the choice of some artificial parameters. While these methods typically provide robust estimates, they assume nominal covariance matrices and thus lack adaptability to unknown and possibly time-varying noise statistical characteristics. In the context of minimum mean square error estimation, heavy-tailed measurement noise modeling with *elliptical* distributions (ED) [14], [15] (such as *Student's t* (ST) [16], [17], [18]) typically employs a Gaussian hierarchical form, introducing identifiability issues between covariance matrices and scale factors [15]. Consequently, accurate estimation for these variables is not guaranteed. Alternatively, Bernoulli mixtures are used for measurement likelihood modeling. However, those combining Gaussian and ED [19] share hierarchical model weaknesses, while Bernoulli latent variable models [20] only detect measurement outliers without adapting to unknown noise statistical characteristics. Some work has focused on filters that are both robust to outliers and adaptive to unknown and possibly time-varying noise statistics [21], [22], [23], [24]. In [21], an ST-distribution-based fixed-interval smoother was proposed but lacks online estimation capability. To address this, [22] introduced a sliding window filter for online noise statistics estimation and joint outlier detection. However, it only uses the current measurement for state trajectory estimation. In addition, [23] introduced a filter using a Gaussian-Gaussian scale mixture (GGSM) distribution, which models inliers with a Gaussian and outliers with a Gaussian scale mixture governed by a Bernoulli variable. This filter was later extended to a distributed framework in [24].

The present work aims at generalizing the variational Bayesian moving-horizon estimation (VB-MHE) approach to accommodate multiple sources of uncertainty in measurement data, including outliers. In the proposed framework, the VB approach is used to directly approximate the joint posterior of state trajectory and noise parameters (e.g., PNCM, MNCM, outlier probability) over a moving window of appropriate length by leveraging all measurements within this window. The proposed VB-MHE framework is flexible enough to incorporate different measurement noise models, including heavy-tailed ED such as ST distribution, as well as multi-model distributions like the Bernoulli-Gaussian (BG) model. For instance, the BG model can be employed specifically to account for potential outliers in the data. This model uses a Bernoulli variable to switch, according to some outlier probability (or rate), between two different Gaussian-distributed processes for outlier and, respectively, inlier measurement noise. This allows for great flexibility, as it decouples inlier and outlier measurements and the respective inlier/outlier noise covariances (INCM/ONCM), unlike models such as the GGSM [23], which adopt a single scalar factor to relate INCM and ONCM. In addition to offering enhanced flexibility in measurement noise modeling, the MHE paradigm enables direct estimation of PNCM instead of PECM by considering the state trajectory across a window rather than a single state. Furthermore, this approach also facilitates improved outlier detection and estimation of the outlier probability. Specifically, MHE defers decision-making, allowing for the reevaluation of past measurements as new data becomes available. Consequently, all estimates for state and noise parameters can be adjusted and refined until a fixed time frame (the window length) elapses. This capability is particularly important in the presence of outliers, where new data can be critical for correctly classifying previous measurements as inliers or outliers. To the best of the authors' knowledge, the ability to infer an unknown, time-varying outlier probability represents a unique feature of the robust adaptive filters developed in this work. This feature is of great practical relevance in scenarios where the occurrence rate of outliers may vary due to environmental changes.

After the derivation of a robust adaptive filter, referred to as R-VB-MHE-BG, for a centralized multi-sensor network wherein measurements from all sensors are conveyed to the filter, we consider the extension to a distributed network wherein each sensor node runs a local filter with the available measurements and can exchange data with the neighbors. To this end, we employed the distributed variational Bayesian (DVB) approach [25] and developed a distributed robust adaptive filter, referred to as R-DVB-MHE-BG. Compared to the hybrid consensus approach adopted for the distributed adaptive filter in [9], which performs consensus iterations at the end of VB iterations, the DVB method exploited here performs a consensus step after each VB iteration, thus allowing a better spread of the consensus outcome throughout the network. As a result, methods based on hybrid consensus might be less effective to ascertain whether sudden changes in measurements are due to PNCM variations or to outliers. On the other hand, the DVB-based approach can easily distinguish these two cases after multiple iterations.

In summary, this paper provides the following main contributions.

1) We propose a general variational Bayesian moving-horizon estimation framework that enables online joint estimation of the state trajectory and noise variables (e.g., PNCM, MNCM and outlier probability) over a moving window.
2) We present detailed derivations of the resulting robust adaptive filter when a Bernoulli-Gaussian (BG) model of the measurement noise is employed, allowing the Bayesian inference process to account for outliers as well as unknown covariances for both normal (inlier) and outlier measurements.
3) We extend the developed robust adaptive filter to a distributed setting by proposing a computationally efficient implementation that leverages the distributed variational Bayesian approach and consensus mechanisms.

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4) We discuss how to incorporate heavy-tailed elliptical distributions, such as ST, into the proposed VB-MHE framework.

The rest of the paper is organized as follows. Section II introduces notation and the problem formulation. Section III derives the centralized robust adaptive filter, referred to as R-VB-MHE-BG. Section IV extends the filter to the case of a distributed sensor network. Section V deals with the case of ED for measurement noise modeling. Section VI provides simulation results to demonstrate the effectiveness of the proposed algorithms. Finally, Section VII ends the paper.

II. Problem Formulation

A. Notation

Given a continuous/discrete random variable v, $v \sim p_v(\cdot)$ indicates that v is distributed according to the *probability density function* (PDF)/ *probability mass function (PMF)* $p_v(\cdot)$. Four distributions will be considered in this paper: Gaussian, inverse Wishart, Beta and Bernoulli. Specifically: $v \sim \mathcal{N}(\cdot; m, \Sigma)$ means that vector $v \in \mathbb{R}^n$ is Gaussian distributed with mean m and covariance Σ; $\Sigma \sim \mathcal{W}^{-1}(\cdot; \Psi, \nu)$ means that the positive definite matrix $\Sigma \in \mathbb{R}^{n\times n}$ is inverse-Wishart distributed with scale matrix $\Psi > 0$ and degree of freedom (DOF) $\nu > n-1$; $\eta \sim \mathfrak{B}(\cdot; \pi)$ means that the binary variable $\eta \in \{0,1\}$ is Bernoulli distributed with probability $\pi \in [0,1]$; $\pi \sim \mathcal{B}(\cdot; e, f)$ means that $\pi \in [0,1]$ is Beta distributed with shape parameters $e > 0$ and $f > 0$. Given two time indices t_1 and t_2 with $t_1 < t_2$, the time window $\{t_1, t_1+1, \ldots, t_2\}$ will be denoted by $t_1 : t_2$. Given the finite set $\mathcal{S}$, the set of variables $\{v^j, j \in \mathcal{S}\}$ will be more simply denoted as $\{v^j\}$.

B. Dynamical and Measurement Model

Consider a linear discrete-time dynamical system

$$x_t = F x_{t-1} + w_t \tag{1}$$

where: $x_t \in \mathbb{R}^{n_x}$ denotes the state vector; F is the state transition matrix; w_t represents a Gaussian zero-mean process disturbance, i.e. $w_t \sim \mathcal{N}(\cdot; 0, Q_t)$, where Q_t is the unknown and possibly time-varying covariance.

The state is observed through measurements $y_t^j \in \mathbb{R}^{n_y^j}$ from multiple agents $j \in \mathcal{S} \triangleq \{1, 2, \ldots, J\}$:

$$y_t^j = H^j x_t + v_t^j \tag{2}$$

with measurement matrices H^j. Each measurement noise v_t^j follows a BG process switching, according to a latent Bernoulli-distributed binary indicator $\eta_t^j \in \{0,1\}$, between the two Gaussian zero-mean processes n_t^j (normal or inlier measurements) and o_t^j (anomalous or outlier measurements), i.e.:

$$\begin{aligned} v_t^j =& (1-\eta_t^j) n_t^j + \eta_t^j o_t^j \\ \eta_t^j \sim& \mathfrak{B}(\cdot; \pi_t^j) \\ n_t^j \sim& \mathcal{N}(\cdot; 0, R_t^j) \\ o_t^j \sim& \mathcal{N}(\cdot; 0, \Sigma_t^j) \end{aligned} \tag{3}$$

where the outlier probability $\pi_t^j = \text{Prob}(\eta_t^j = 1)$, the inlier noise covariance matrix (INCM) R_t^j, and the outlier noise covariance matrix (ONCM) Σ_t^j are assumed to be unknown and possibly time-varying. Additionally, x_0, w_t, n_t^j, o_t^j, and η_t^j are assumed to be mutually uncorrelated.

For a given length $T \geq 0$ of the moving horizon, the problem of interest is to recursively estimate the state trajectory $x_{t-T:t} \triangleq \{x_{t-T}, \ldots, x_t\}$ along with the unknown (possibly time-varying) process noise covariance matrix (PNCM) Q_t, and, for each agent j, INCM R_t^j, ONCM Σ_t^j, the binary indicator η_t^j, the outlier probability π_t^j. We refer to x_t, Q_t in the dynamical model as global variables and, on the other hand, to η_t^j, π_t^j, R_t^j, Σ_t^j in the measurement model as local variables relative to agent j.

In the standard Kalman filter framework, explicitly reconstructing the probability density function (PDF) of the PNCM is a challenging problem. To address this, we propose a moving horizon estimation approach, incorporating the PDF of Q_t into the joint PDF. This method enables better estimation of noise statistics by considering the entire state trajectory over a moving window. Since the joint posterior distribution is not Gaussian and a closed-form solution for it cannot be obtained, we adopt a variational Bayesian approach to jointly estimate the system state trajectory and the unknown noise parameters.

Notice that to develop a robust filter accounting for heavy-tailed distributions and/or outliers, several models can be employed to characterize the measurement noise v_t^j. To illustrate the proposed VB-MHE framework, we focus on the BG model (3), which can effectively handle outliers as shown hereafter. Later, in Section V, we will discuss how different measurement noise models can be incorporated into the proposed VB-MHE framework.

III. Outlier-Robust Centralized Variational Bayesian Moving Horizon Estimation

This section deals with the joint estimation of the state trajectory $x_{t-T:t}$ along with all the noise parameters by collecting measurements from all sensors within the time window $t-T+1:t$. Hereafter, covariances Q, $\{R^j\}$ and outlier probabilities $\{\pi^j\}$ are modeled as constant within the moving window, though they can vary from one window to another. As a result, time subscripts are not used for these parameters during the filter derivation. This choice, which is instrumental to simplify computations, is reasonable whenever the rate of variation of these parameters is slow compared to the window size (see, for instance, the scenario considered in Section VI).

It is assumed that the initial state x_0 follows a Gaussian distribution with mean and covariance $(\bar{x}_0, \bar{P}_0)$ and that the initial unknown noise covariances Q and R^j have IW priors with parameters $(\bar{M}_0, \bar{m}_0)$ and $(\bar{S}_0^j, \bar{s}_0^j)$, respectively. Additionally, the initial outlier probability has a Beta prior with parameters $(\bar{e}_0^j, \bar{f}_0^j)$. Then, the joint prior PDF at time $t=0$ is given by

$$\begin{aligned} p\left(x_0, Q, \{R^j\}, \{\pi^j\}\right) =& \mathcal{N}(x_0; \bar{x}_0, \bar{P}_0)\, \mathcal{W}^{-1}(Q; \bar{M}_0, \bar{m}_0) \\ & \times \prod_{j\in\mathcal{S}} \mathcal{W}^{-1}(R^j; \bar{S}_0^j, \bar{s}_0^j)\, \mathcal{B}(\pi^j; \bar{e}_0^j, \bar{f}_0^j). \end{aligned} \tag{4}$$

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A. Variational Bayesian Measurement Update

Operating on a moving-horizon window of length T, the robust VB-MHE (R-VB-MHE) filter supposes that the information collected up to time $t-T$ is approximately represented by the following joint prior PDF of $\{x_{t-T}, Q, \{R^j\}, \{\pi^j\}\}$:

$$\begin{aligned} &p\left(x_{t-T}, Q, \{R^j\}, \{\pi^j\} \mid \{y^j_{1:t-T}\}\right) \\ &\propto \mathcal{N}(x_{t-T}; \bar{x}_{t-T}, \bar{P}_{t-T})\, \mathcal{W}^{-1}(Q; \bar{M}_{t-T}, \bar{m}_{t-T}) \\ &\quad \times \prod\nolimits_{j\in\mathcal{S}} \mathcal{W}^{-1}(R^j; \bar{S}^j_{t-T}, \bar{s}^j_{t-T})\, \mathcal{B}(\pi^j; \bar{e}^j_{t-T}, \bar{f}^j_{t-T}). \end{aligned} \tag{5}$$

Subsequently, measurements collected over the window $t-T+1:t$ are used to update the posterior PDF of the joint variable $\Theta_{t-T:t} \triangleq \{x_{t-T:t}, Q, \{R^j\}, \{\eta^j_{t-T+1:t}\}, \{\pi^j\}, \{\Sigma^j_{t-T+1:t}\}\}$. The likelihood in the moving window $t-T+1:t$ under the BG model (3) is given by:

$$\begin{aligned} &p\Big(\{y^j_{t-T+1:t}\} | x_{t-T+1:t}, \{\eta^j_{t-T+1:t}\}, \{\Sigma^j_{t-T+1:t}\}, \{R^j\}\Big) \\ &= \prod_{j\in\mathcal{S}} \prod_{i=t-T+1}^{t} \mathcal{N}(y^j_i; H^j x_i, R^j)^{1-\eta^j_i}\, \mathcal{N}(y^j_i; H^j x_i, \Sigma^j_i)^{\eta^j_i}. \end{aligned} \tag{6}$$

The prior distribution of the state trajectory within the window $t-T+1:t$ can be derived using the dynamical system (1) as

$$p(x_{t-T+1:t} | \{y^j_{1:t-T}\}) = \prod_{i=t-T+1}^{t} \mathcal{N}(x_i; F x_{i-1}, Q). \tag{7}$$

Additionally, the prior distribution of the outlier indicators $\{\eta^j_{t-T+1:t}\}$ and ONCMs $\{\Sigma^j_{t-T+1:t}\}$ over the window $t-T+1:t$ is given by:

$$\begin{aligned} p\left(\{\eta^j_{t-T+1:t}\}, \{\Sigma^j_{t-T+1:t}\}\right) &= \prod_{j\in\mathcal{S}} \prod_{i=t-T+1}^{t} \mathfrak{B}(\eta^j_i; \pi^j) \\ &\quad \times \mathcal{W}^{-1}(\Sigma^j_i; \bar{U}^j_0, \bar{u}^j_0). \end{aligned} \tag{8}$$

Notice that we suppose the outlier covariances Σ^j_i at different time instants to be completely independent, reflecting the fact that outliers are generally completely unpredictable and do not provide informative data. Accordingly, we avoid using any memory in the estimation of the outlier covariances by considering, in (8), priors $\mathcal{W}^{-1}(\Sigma^j_i; \bar{U}^j_0, \bar{u}^j_0)$ with fixed parameters $\bar{U}^j_0$ and $\bar{u}^j_0$ for any time $i = t-T+1, \ldots, t$.

Given the prior knowledge summarized by $\bar{x}_{t-T}$, $\bar{P}_{t-T}$, $\bar{M}_{t-T}$, $\bar{m}_{t-T}$, $\{\bar{S}^j_{t-T}\}$, $\{\bar{s}^j_{t-T}\}$, $\{\bar{e}^j_{t-T}\}$, $\{\bar{f}^j_{t-T}\}$ in (5) and incorporating the likelihood in (6), the prior PDF of the state trajectory $x_{t-T+1:t}$ in (7), and the prior PDF of $\{\eta^j_{t-T+1:t}\}$ and $\{\Sigma^j_{t-T+1:t}\}$ in (8) with parameters $\{\pi^j\}$, $\{\bar{U}^j_0\}$ and $\{\bar{u}^j_0\}$, the aim is to approximate the posterior PDF of $\Theta_{t-T:t}$ using the VB approach to characterize the joint distribution across all agents and time instants within the moving horizon. To better capture the conditional independence between random variables, we present a graphical representation of the model in Fig. 1.

The goal is to obtain the joint posterior PDF $p(\Theta_{t-T:t} \mid \{y^j_{1:t}\})$, according to Bayes rule, given the set of measurements

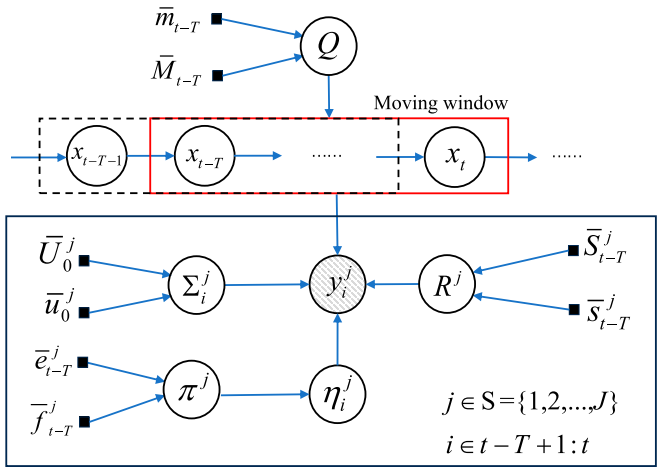


Fig. 1. Graphical model of state and noise variables over a moving window.

$\{y^j_{t-T+1:t}\}$ collected over the moving horizon from all agents $j \in \mathcal{S}$ and the prior PDFs in (5), (7), and (8), as

$$p(\Theta_{t-T:t} \mid \{y^j_{1:t}\}) = \frac{p(\Theta_{t-T:t}, \{y^j_{t-T+1:t}\} \mid \{y^j_{1:t-T}\})}{p(\{y^j_{t-T+1:t}\} \mid \{y^j_{1:t-T}\})}. \tag{9}$$

The numerator in (9) is the joint PDF of $\Theta_{t-T:t}$ and $\{y^j_{t-T+1:t}\}$, given the joint prior in (5), the prior of the state trajectory $x_{t-T+1:t}$ in (7), and the joint prior of $\{\Sigma^j_{t-T+1:t}\}$ and $\{\eta^j_{t-T+1:t}\}$ in (8), which turns out to be

$$\begin{aligned} &p(\Theta_{t-T:t}, \{y^j_{t-T+1:t}\} \mid \{y^j_{1:t-T}\}) \\ &\propto \prod_{j\in\mathcal{S}} \prod_{i=t-T+1}^{t} \mathcal{N}(y^j_i; H^j x_i, R^j)^{1-\eta^j_i}\, \mathcal{N}(y^j_i; H^j x_i, \Sigma^j_i)^{\eta^j_i} \\ &\times \left[\prod_{i=t-T+1}^{t} \mathcal{N}(x_i; F x_{i-1}, Q)\right] \mathcal{N}(x_{t-T}; \bar{x}_{t-T}, \bar{P}_{t-T}) \\ &\times \mathcal{W}^{-1}(Q; \bar{M}_{t-T}, \bar{m}_{t-T}) \\ &\times \prod_{j\in\mathcal{S}} \prod_{i=t-T+1}^{t} \mathfrak{B}(\eta^j_i; \pi^j) \mathcal{W}^{-1}(\Sigma^j_i; \bar{U}^j_0, \bar{u}^j_0) \\ &\times \prod_{j\in\mathcal{S}} \left[\mathcal{B}(\pi^j; \bar{e}^j_{t-T}, \bar{f}^j_{t-T})\, \mathcal{W}^{-1}(R^j; \bar{S}^j_{t-T}, \bar{s}^j_{t-T})\right] \end{aligned} \tag{10}$$

and the denominator in (9) is the model evidence [18] that, due to the non-Gaussian nature of all variables in $\Theta_{t-T:t} \setminus \{x_{t-T:t}\}$, cannot be obtained in closed form. Moreover, using the standard Bayes rule, the posterior (9) is no longer of the same factored form of the prior (5) (lack of prior conjugacy). Hence, the VB approach is used to approximate the posterior $p(\Theta_{t-T:t} \mid \{y^j_{1:t}\})$ by a variational PDF $q(\Theta_{t-T:t})$ of the form

$$\begin{aligned} q(\Theta_{t-T:t}) &= \prod_{\theta\in\Theta_{t-T:t}} q_\theta(\theta) = q_x(x_{t-T:t})\, q_Q(Q) \\ &\times \prod_{j\in\mathcal{S}} \left\{ q_{R^j}(R^j)\, q_{\pi^j}(\pi^j) \prod_{i=t-T+1}^{t} \left[q_{\eta^j_i}(\eta^j_i)\, q_{\Sigma^j_i}(\Sigma^j_i) \right] \right\} \end{aligned} \tag{11}$$

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which is found by minimizing the Kullback-Leibler (KL) divergence defined as follows:

$$\begin{aligned}&\mathrm{KL}(q(\Theta_{t-T:t})||p(\Theta_{t-T:t}\mid\{y^j_{1:t}\}))\\&=\int q(\Theta_{t-T:t})\ln\frac{q(\Theta_{t-T:t})}{p(\Theta_{t-T:t}\mid\{y^j_{1:t}\})}\,d\Theta_{t-T:t}\\&=-\mathbb{E}_{\Theta_{t-T:t}}\left[\ln\frac{p(\Theta_{t-T:t},\{y^j_{t-T+1:t}\}\mid\{y^j_{1:t-T}\})}{q(\Theta_{t-T:t})}\right]\\&\quad+\ln p(\{y^j_{t-T+1:t}\}\mid\{y^j_{1:t-T}\})\\&=-\mathcal{L}(q(\Theta_{t-T:t}))+\ln p(\{y^j_{t-T+1:t}\}\mid\{y^j_{1:t-T}\})\end{aligned}\tag{12}$$

where $\mathcal{L}(q(\Theta_{t-T:t}))$ represents the evidence lower bound (ELBO) defined as

$$\begin{aligned}&\mathcal{L}(q(\Theta_{t-T:t}))\\&\quad=\mathbb{E}_{\Theta_{t-T:t}}\left[\ln\frac{p(\Theta_{t-T:t},\{y^j_{t-T+1:t}\}\mid\{y^j_{1:t-T}\})}{q(\Theta_{t-T:t})}\right].\end{aligned}\tag{13}$$

Since $\ln p(\{y^j_{t-T+1:t}\}\mid\{y^j_{1:t-T}\})$ does not depend on $\Theta_{t-T:t}$, minimizing (12) is equivalent to maximizing the ELBO (13). The optimal solution

$$q^*(\Theta_{t-T:t})=\arg\max_{q(\Theta_{t-T:t})}\mathcal{L}(q(\Theta_{t-T:t}))\tag{14}$$

can be found via the mean-field VB inference approach [26] as follows

$$q^*(\theta)=\arg\max_{q(\theta)}\mathcal{L}(q^*(\Theta_{t-T:t}\setminus\{\theta\}),q(\theta))\tag{15}$$

where $q^*(\theta)$, used as shorthand notation for $q^*_\theta(\theta)$, represents the approximate posterior PDF/PMF of any variable $\theta\in\Theta_{t-T:t}$ in (11).

By maximizing the ELBO using the *coordinate ascent* method [26], the optimal solution $q^*(\theta)$ in (15) satisfies

$$\begin{aligned}\ln q^*(\theta)=\mathbb{E}_{\Theta_{t-T:t}\setminus\{\theta\}}\left[\ln p(\Theta_{t-T:t},\{y^j_{t-T+1:t}\}\mid\{y^j_{1:t-T}\})\right]\\+c_\theta\end{aligned}\tag{16}$$

where c_θ denotes a constant with respect to θ.

The following propositions provide the expressions for each of the variational factors involved in the VB optimal solution (16).

Proposition 1: Given the joint PDF (10) and the factorized approximation (11), the approximate PDF of the state trajectory according to the VB approach is of the Gaussian form

$$q^*(x_{t-T:t})=\mathcal{N}(x_{t-T:t};\hat{x}_{t-T:t},P_{t-T:t})\tag{17}$$

where

$$\hat{x}_{t-T:t}=\Omega^{-1}_{t-T:t}\,\omega_{t-T:t}\tag{18}$$

$$P_{t-T:t}=\Omega^{-1}_{t-T:t}\tag{19}$$

$$\omega_{t-T:t}=\begin{bmatrix}\sum_{j\in\mathcal{S}}(H^j)'\Psi^j_t y^j_t\\\vdots\\\sum_{j\in\mathcal{S}}(H^j)'\Psi^j_{t-T+1}y^j_{t-T+1}\\\bar{P}^{-1}_{t-T}\,\bar{x}_{t-T}\end{bmatrix}\tag{20}$$

$$\Omega_{t-T:t}=\begin{bmatrix}\Omega^-&\Omega^\#&0&\cdots&0\\\Omega^\times&\Omega^*_{t-1}&\Omega^\#&&\vdots\\0&\Omega^\times&\ddots&\ddots&\vdots\\\vdots&&\ddots&\Omega^*_{t-T+1}&\Omega^\#\\0&\cdots&\cdots&\Omega^\times&\Omega^+\end{bmatrix}\tag{21}$$

$$\Psi^j_i=(1-\mathbb{E}[\eta^j_i])\,\mathbb{E}[(R^j)^{-1}]+\mathbb{E}[\eta^j_i]\,\mathbb{E}[(\Sigma^j_i)^{-1}]\tag{22}$$

with the following sub-elements in (21)

$$\begin{aligned}\Omega^-&=\sum_{j\in\mathcal{S}}(H^j)'\Psi^j_tH^j+\Phi\\\Omega^\#&=-\Phi F\\\Omega^\times&=-F'\Phi\\\Omega^*_i&=\sum_{j\in\mathcal{S}}(H^j)'\Psi^j_iH^j+\Phi+F'\Phi F\\\Omega^+&=F'\Phi F+(\bar{P}_{t-T})^{-1}\end{aligned}\tag{23}$$

$$\Phi=\mathbb{E}[Q^{-1}]\tag{24}$$

Proof: See Appendix B in the supplementary material. □

Notice that $\hat{x}_{t-T:t}$ and $P_{t-T:t}$ in (18)-(19) have the following block-partitioned structure

$$\hat{x}_{t-T:t}=\left[\;\hat{x}'_t,\hat{x}'_{t-1},\cdots,\hat{x}'_{t-T}\;\right]'\tag{25}$$

$$P_{t-T:t}=\begin{bmatrix}P_t&P_{t,t-1}&\cdots&P_{t,t-T}\\P_{t-1,t}&P_{t-1}&\cdots&P_{t-1,t-T}\\\vdots&\vdots&\ddots&\vdots\\P_{t-T,t}&P_{t-T,t-1}&\cdots&P_{t-T}\end{bmatrix}.\tag{26}$$

that will be useful for the subsequent developments.

Proposition 2: Given the joint PDF (10) and the factorized approximation (11), the approximate PDF of the PNCM according to the VB approach is of the IW form

$$q^*(Q)=\mathcal{W}^{-1}(Q;M_t,m_t)\tag{27}$$

where the PNCM scale matrix M_t and DOF m_t in (27) are computed by

$$\begin{aligned}m_t&=\bar{m}_{t-T}+T\\M_t&=\bar{M}_{t-T}+\sum_{i=t-T+1}^{t}[(\hat{x}_i-F\hat{x}_{i-1})(\hat{x}_i-F\hat{x}_{i-1})'\\&\quad+P_i+FP_{i-1}F'-P_{i,i-1}F'-FP_{i-1,i}]\end{aligned}\tag{28}$$

and

$$\mathbb{E}[Q^{-1}]=m_tM_t^{-1}.\tag{29}$$

Proof: See Appendix C in the supplementary material. □

Proposition 3: Given the joint PDF (10) and the factorized approximation (11), the approximate PDF of the outlier probability π^j is of the Beta form

$$q^*(\pi^j)=\mathcal{B}(\pi^j;e^j_t,f^j_t)\tag{30}$$

with shape parameters

$$e^j_t=\bar{e}^j_{t-T}+\sum_{i=t-T+1}^{t}\mathbb{E}[\eta^j_i]$$

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$$f_t^j = \bar{f}_{t-T}^j + \sum_{i=t-T+1}^{t} (1 - \mathbb{E}[\eta_i^j]) \tag{31}$$

and

$$\mathbb{E}[\pi^j] = \frac{e_t^j}{e_t^j + f_t^j}. \tag{32}$$

Proof: See Appendix D in the supplementary material. □

Proposition 4: Given the joint PDF (10) and the factorized approximation (11), the approximate PDF of the outlier indicator η_i^j is of the Bernoulli form

$$q^*(\eta_i^j) = \mathfrak{B}(\eta_i^j; \xi_i^j) \tag{33}$$

where $\xi_i^j = \text{Prob}(\eta_i^j = 1)$ is given by

$$\xi_i^j = \frac{\alpha_i^j}{\alpha_i^j + \beta_i^j} \tag{34}$$

and

$$\mathbb{E}[\eta_i^j] = \xi_i^j. \tag{35}$$

In (34), α_i^j and β_i^j are given by

$$\begin{aligned}\alpha_i^j &= \exp\left(\mathbb{E}[\ln \pi^j] - \frac{1}{2}\operatorname{tr}(D_i^j\, \mathbb{E}[(\Sigma_i^j)^{-1}]) - \frac{1}{2}\mathbb{E}[\ln |\Sigma_i^j|]\right)\\ \beta_i^j &= \exp\left(\mathbb{E}[\ln(1-\pi^j)] - \frac{1}{2}\operatorname{tr}(D_i^j\, \mathbb{E}[(R^j)^{-1}])\right.\\ &\qquad \left. -\frac{1}{2}\mathbb{E}[\ln |R^j|]\right)\end{aligned} \tag{36}$$

with

$$D_i^j = (y_i^j - H^j \hat{x}_i)(y_i^j - H^j \hat{x}_i)' + H^j P_i (H^j)' \tag{37}$$

and

$$\begin{aligned}\mathbb{E}[\ln \pi^j] &= \psi(e_t^j) - \psi(e_t^j + f_t^j)\\ \mathbb{E}[\ln(1-\pi^j)] &= \psi(f_t^j) - \psi(e_t^j + f_t^j)\\ \mathbb{E}[\ln |R^j|] &= \ln |S_t^j| - n_y^j \ln 2 - \sum_{k=1}^{n_y^j} \psi\left(\frac{1}{2}(s_t^j + 1 - k)\right)\\ \mathbb{E}[\ln |\Sigma_i^j|] &= \ln |U_i^j| - n_y^j \ln 2 - \sum_{k=1}^{n_y^j} \psi\left(\frac{1}{2}(u_i^j + 1 - k)\right).\end{aligned} \tag{38}$$

In (38), $\mathbb{E}[\ln |R^j|]$ and $\mathbb{E}[\ln |\Sigma_i^j|]$ denote the expectations of the log-determinant of an IW matrix [27], and $\psi(\cdot)$ denotes the digamma function [27].

Proof: See Appendix E in the supplementary material. □

Proposition 5: Given the joint PDF (10) and the factorized approximation (11), the approximate PDF of the outlier covariance according to the VB approach is of the IW form

$$q^*(\Sigma_i^j) = \mathcal{W}^{-1}(\Sigma_i^j; U_i^j, u_i^j) \tag{39}$$

where the OCNM scale matrix U_i^j and DOF u_i^j in (39) are computed as follows

$$u_i^j = \bar{u}_0^j + \mathbb{E}[\eta_i^j]$$

$$U_i^j = \bar{U}_0^j + \mathbb{E}[\eta_i^j]\, D_i^j \tag{40}$$

with D_i^j given by (37). Then, the expectation of the inverse ONCM is given by

$$\mathbb{E}[(\Sigma_i^j)^{-1}] = u_i^j\, (U_i^j)^{-1}. \tag{41}$$

Proof: See Appendix F in the supplementary material. □

Proposition 6: Given the joint posterior (10) and the factorized approximation (11), the approximate PDF of the inlier covariance according to the VB approach is of the IW form

$$q^*(R^j) = \mathcal{W}^{-1}(R^j; S_t^j, s_t^j) \tag{42}$$

where the INCM scale matrix S_t^j and DOF s_t^j in (42) are computed by

$$\begin{aligned}s_t^j &= \bar{s}_{t-T}^j + \sum_{i=t-T+1}^{t} \left(1 - \mathbb{E}[\eta_i^j]\right)\\ S_t^j &= \bar{S}_{t-T}^j + \sum_{i=t-T+1}^{t} \left(1 - \mathbb{E}[\eta_i^j]\right)\, D_i^j\end{aligned} \tag{43}$$

with D_i^j given by (37). Then, the expectation of the inverse INCM is given by

$$\mathbb{E}[(R^j)^{-1}] = s_t^j\, (S_t^j)^{-1}. \tag{44}$$

Proof: See Appendix G in the supplementary material. □

Due to the coupling of the variational variables of $q^*(\theta)$ for $\theta = x_{t-T:t}, Q, \eta_i^j, \pi^j, \Sigma_i^j, R^j$, the above Propositions 1-6 provide a set of coupled nonlinear equations in the unknowns $\hat{x}_{t-T:t}$, $P_{t-T:t}$, M_t, m_t, S_t^j, s_t^j, U_i^j, u_i^j, e_t^j, f_t^j, ξ_i^j, whose solution can be addressed through the variational fixed-point method [4], [8]. In this iterative method, the optimal variational PDF $q^*(\theta)$ of any given element θ can be updated iteratively by simply fixing all the other factors in $\Theta_{t-T:t} \setminus \{\theta\}$, ensuring convergence towards a local optimum. The resulting (centralized) R-VB-MHE filter, using N (for $n = 0, 1, \ldots, N-1$) VB iterations, is summarized in Algorithm 1.

B. Time-Propagation to the Next Window

The time-propagation of PNCM, INCM, and outlier probability takes the following form

$$\begin{aligned}\bar{M}_{t-T+1} &= \rho M_t^N\\ \bar{m}_{t-T+1} &= \rho(m_t^N - n_x - 1) + n_x + 1\\ \bar{S}_{t-T+1}^j &= \rho S_t^{j,N}\\ \bar{s}_{t-T+1}^j &= \rho(s_t^{j,N} - n_y^j - 1) + n_y^j + 1\\ \bar{e}_{t-T+1}^j &= \rho e_t^{j,N}\\ \bar{f}_{t-T+1}^j &= \rho f_t^{j,N}\end{aligned} \tag{45}$$

as in [8], where $\rho \in (0, 1]$ is a forgetting factor. Moreover, for the state time propagation from the current window to the next window, we have

$$\begin{aligned}\tilde{x}_{t-T+1} &= F \bar{x}_{t-T}\\ \hat{Q}_{t-T+1} &= \mathbb{E}[Q] = \frac{\bar{M}_{t-T+1}}{\bar{m}_{t-T+1} - n_x - 1}\end{aligned}$$

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Algorithm 1: R-VB-MHE-BG (window $t-T:t$)

Inputs: $\bar{x}_{t-T}$, $\bar{P}_{t-T}$, F, $\{H^j\}$, $\{y^j_{t-T+1:t}\}$, $\bar{M}_{t-T}$, $\bar{m}_{t-T}$, $\{\bar{S}^j_{t-T}\}$, $\{\bar{s}^j_{t-T}\}$, $\{\bar{U}^j_0\}$, $\{\bar{u}^j_0\}$, $\{\bar{e}^j_{t-T}\}$, $\{\bar{f}^j_{t-T}\}$, N, T, ρ

VB initialization:

for each $j \in \mathcal{S}$ and $i \in t-T+1:t$

$\mathbb{E}^0[\eta^j_i] = 0$, $\mathbb{E}^0[\ln \pi^j] = \psi(\bar{e}^j_{t-T}) - \psi(\bar{e}^j_{t-T} + \bar{f}^j_{t-T})$,

$\mathbb{E}^0[\ln(1-\pi^j)] = \psi(\bar{f}^j_{t-T}) - \psi(\bar{e}^j_{t-T} + \bar{f}^j_{t-T})$,

$\mathbb{E}^0[(Q)^{-1}] = \bar{m}_{t-T}(\bar{M}_{t-T})^{-1}$, $\mathbb{E}^0[(R^j)^{-1}] = \bar{s}^j_{t-T}(\bar{S}^j_{t-T})^{-1}$,

$\mathbb{E}^0[(\Sigma^j_i)^{-1}] = \bar{u}^j_0(\bar{U}^j_0)^{-1}$,

$\mathbb{E}^0[\ln|R^j|] = \ln|\bar{S}^j_{t-T}| - n^j_y \ln 2 - \sum_{k=1}^{n^j_y} \psi(0.5(\bar{s}^j_{t-T}+1-k))$,

$\mathbb{E}^0[\ln|\Sigma^j_i|] = \ln|\bar{U}^j_0| - n^j_y \ln 2 - \sum_{k=1}^{n^j_y} \psi(0.5(\bar{u}^j_0+1-k))$;

Set $m_t = \bar{m}_{t-T} + T$;

VB-MH measurement update for any $j \in \mathcal{S}$

for $n = 0, 1, \ldots, N-1$ do

1: Compute $\Psi^{j,n}_i$ using $\mathbb{E}^n[\eta^j_i]$, $\mathbb{E}^n[(R^j)^{-1}]$, and $\mathbb{E}^n[(\Sigma^j_i)^{-1}]$ for each $i \in t-T+1:t$ via (22);

2: Compute Φ^n using $\mathbb{E}^n[Q^{-1}]$ via (24);

3: Compute $\omega^{n+1}_{t-T:t}$ using $\Psi^{j,n}_{t-T+1:t}$ via (20);

4: Compute $\Omega^{n+1}_{t-T:t}$ using $\Psi^{j,n}_{t-T+1:t}$ and Φ^n via (21) and (23);

5: Compute $\hat{x}^{n+1}_{t-T:t}$ and $P^{n+1}_{t-T:t}$ using $\Omega^{n+1}_{t-T:t}$ and $\omega^{n+1}_{t-T:t}$ via (18)-(19);

6: Compute M^{n+1}_t using $P^{n+1}_{t-T:t}$ and $\hat{x}^{n+1}_{t-T:t}$ via (28);

7: Compute $\mathbb{E}^{n+1}[Q^{-1}]$ using M^{n+1}_t and m_t via (29);

for $i = t-T+1, t-T+2, \cdots, t$ do

8: Compute $D^{j,n+1}_i$ using P^{n+1}_i and $\hat{x}^{n+1}_i$ via (37);

9: Compute $\mathbb{E}^{n+1}[\eta^j_i]$ using $\mathbb{E}^n[\ln \pi^j]$, $\mathbb{E}^n[(\Sigma^j_i)^{-1}]$, $\mathbb{E}^n[\ln|\Sigma^j_i|]$, $\mathbb{E}^n[\ln(1-\pi^j)]$, $\mathbb{E}^n[(R^j)^{-1}]$, $\mathbb{E}^n[\ln|R^j|]$, and $D^{j,n+1}_i$ via (34)-(36);

10: Compute $U^{j,n+1}_i$, $u^{j,n+1}_i$ using $\mathbb{E}^{n+1}[\eta^j_i]$, $D^{j,n+1}_i$ via (40);

11: Compute $\mathbb{E}^{n+1}[(\Sigma^j_i)^{-1}]$ using $U^{j,n+1}_i$ and $u^{j,n+1}_i$ via (41);

12: Compute $\mathbb{E}^{n+1}[\ln|\Sigma^j_i|]$ using $U^{j,n+1}_i$ and $u^{j,n+1}_i$ via (38);

end for

13: Compute $e^{j,n+1}_t$ and $f^{j,n+1}_t$ using $\mathbb{E}^{n+1}[\eta^j_i]$ for $i = t-T+1:t$ via (31);

14: Compute $\mathbb{E}^{n+1}[\pi^j]$, $\mathbb{E}^{n+1}[\ln \pi^j]$, and $\mathbb{E}^{n+1}[\ln(1-\pi^j)]$ using $e^{j,n+1}_t$ and $f^{j,n+1}_t$ via (32) and (38);

15: Compute $S^{j,n+1}_t$ and $s^{j,n+1}_t$ using $\mathbb{E}^{n+1}[\eta^j_i]$ and $D^{j,n+1}_i$ for $i = t-T+1:t$ via (43);

16: Compute $\mathbb{E}^{n+1}[(R^j)^{-1}]$ using $S^{j,n+1}_t$ and $s^{j,n+1}_t$ via (44);

17: Compute $\mathbb{E}^{n+1}[\ln|R^j|]$ using $S^{j,n+1}_t$ and $s^{j,n+1}_t$ via (38);

end for

Moving horizon time-propagation for any $j \in \mathcal{S}$

Update $\bar{M}_{t-T+1}, \bar{m}_{t-T+1}, \bar{S}^j_{t-T+1}, \bar{s}^j_{t-T+1}, \bar{e}^j_{t-T+1}, \bar{f}^j_{t-T+1}$ via (45);

Compute $\bar{x}_{t-T+1}$, $\bar{P}_{t-T+1}$, and $\hat{Q}_{t-T+1}$ via (46)-(47);

Compute $\hat{R}^j_{t-T+1} = \mathbb{E}[R^j] = \frac{\bar{S}^j_{t-T+1}}{\bar{s}^j_{t-T+1} - n^j_y - 1}$;

Outputs: $\bar{x}_{t-T+1}$, $\bar{P}_{t-T+1}$, $\bar{M}_{t-T+1}$, $\bar{m}_{t-T+1}$, $\{\bar{S}^j_{t-T+1}\}$, $\{\bar{s}^j_{t-T+1}\}$, $\hat{Q}_{t-T+1}$, $\{\hat{R}^j_{t-T+1}\}$, $\{\bar{e}^j_{t-T+1}\}$, $\{\bar{f}^j_{t-T+1}\}$, $\{\mathbb{E}^N[\pi^j]\}$

$$\begin{aligned}
\tilde{P}_{t-T+1} &= F\bar{P}_{t-T}F' + \hat{Q}_{t-T+1} \\
\bar{P}_{t-T+1} &= (\tilde{P}^{-1}_{t-T+1} + \Omega^N_{t-T+1})^{-1} \\
\bar{x}_{t-T+1} &= \bar{P}_{t-T+1}(\tilde{P}^{-1}_{t-T+1}\tilde{x}_{t-T+1} + \omega^N_{t-T+1})
\end{aligned} \tag{46}$$

where Ω^N_{t-T+1} and ω^N_{t-T+1} take the following form

$$\begin{aligned}
\Omega^N_{t-T+1} &= \sum_{j\in\mathcal{S}} (H^j)' \Psi^{j,N}_{t-T+1} H^j \\
\omega^N_{t-T+1} &= \sum_{j\in\mathcal{S}} (H^j)' \Psi^{j,N}_{t-T+1} y^j_{t-T+1}\,.
\end{aligned} \tag{47}$$

Notice that, according to the proposed MHE framework, when moving from the current time window $t-T:t$ to the next one $t-T+1:t+1$, only the estimates at time $t-T$ are fixed while all the subsequent estimates are reevaluated and possibly modified by taking into account also the new measurements $\{y^j_{t+1}\}$. This is particularly important in the presence of outliers, as the future information might be crucial to distinguish between inliers and outliers, especially when the outlier probability is unknown and has to be estimated.

Remark 1: From (22), we observe that the covariance used in the likelihood calculation depends on the expectation of Bernoulli variable $\mathbb{E}[\eta^j_i]$, which rules the switching between different covariances, specifically R^j and Σ^j_i. Notably, the VB-MHE algorithm in [8] is a special case of the R-VB-MHE-BG algorithm proposed here when $\mathbb{E}[\eta^j_i] = 1$.

Remark 2: Given that outliers are characterized by a much larger covariance than inliers, the prior scale matrix U^j_0 of the ONCM should be set at least one order of magnitude larger than the prior scale matrix S^j_0 of the INCM to distinguish between inlier covariance R_t and outlier covariance Σ_t; for instance, in the simulations of Section VI, we set $U^j_0 = 100S^j_0$. Furthermore, the proposed filter can be extended by imposing covariance bounds using constrained IW distributions, following [8]. Such constraints are particularly useful for distinguishing inliers from outliers and enhancing estimation accuracy.

Remark 3: The proposed filter has been derived under the assumption of slow variations of the noise parameters, so that they can be approximately modeled as constant within each time window $t-T:t$. In case such an assumption does not hold, the approach can be extended by considering time-indexed noise parameters Q_t, $\{R^j_t\}$, $\{\pi^j_t\}$ in (6)-(11) and incorporating Markov transition densities for these parameters in the expression for the joint posterior PDF (10). However, we do not pursue this generalization to avoid overly cumbersome notation.

IV. Outlier-Robust Distributed Variational Bayesian Moving Horizon Estimation

In this section, we consider the same estimation problem formulated in the previous section over a multi-agent network $(\mathcal{S}, \mathcal{A})$ where each agent $i \in \mathcal{S}$ can exchange data according to the communication links in $\mathcal{A} \subseteq \mathcal{S} \times \mathcal{S}$, precisely meaning that $(j,i) \in \mathcal{A}$ if agent i can receive data from agent j. In particular, $\mathcal{S}_i$ will denote the set of in-neighbors (including node i itself) from which agent i can receive data. For such a network, the aim is to extend R-VB-MHE to a completely distributed setting wherein each agent can accomplish the estimation task independently of the others by just processing local data and data from in-neighbors. In the distributed filter, each agent $j \in \mathcal{S}$ locally computes, at each time t, parameter estimates $\hat{x}^j_{t-T:t}$, $P^j_{t-T:t}$, m^j_t, M^j_t for the global variables $x_{t-T:t}$ and Q only based on its local measurements and the information exchanged with its in-neighborhood $\mathcal{S}_j$.

A. Distributed Variational Bayesian Measurement Update

In this section, we are going to exploit the *distributed variational Bayesian* (DVB) approach, originally proposed in [25] for distributed static estimation, in our distributed dynamic estimation problem. First of all, we notice that the global lower

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bound can be rewritten as an averaged local lower bound over sensors as follows:

$$\mathcal{L}(q(\Theta_{t-T:t})) = \frac{1}{J}\sum_{j=1}^{J}\mathcal{L}_j(q(\Theta_{t-T:t})) \tag{48}$$

where the local lower bound $\mathcal{L}_j(q(\Theta_{t-T:t}))$ is defined as

$$\mathcal{L}_j(q(\Theta_{t-T:t})) \triangleq \mathbb{E}_{\Theta_{t-T:t}}\left[\ln \frac{p(y^j_{t-T+1:t} \mid \Theta_{t-T:t}, \{y^j_{1:t-T}\})^J\, p(\Theta_{t-T:t})}{q(\Theta_{t-T:t})}\right]. \tag{49}$$

It is convenient to partition the overall set of variables $\Theta_{t-T:t}$ into the subset of global variables $\Delta_{t-T:t} = \{x_{t-T:t}, Q\}$ shared by all agents and complementary subset $\Xi_{t-T:t} = \bigcup_{j\in\mathcal{S}} \Xi^j_{t-T:t}$ where $\Xi^j_{t-T:t} = \{R^j, \eta^j_{t-T+1:t}, \pi^j, \Sigma^j_{t-T+1:t}\}$ is the set of local variables for agent $j \in \mathcal{S}$. Next, it is worth pointing out that all involved distributions (Gaussian, inverse-Wishart, Beta, Bernoulli) belong to exponential families and can, therefore, be uniquely characterized in terms of their natural parameters. In this respect, let us denote, for each variable $\theta \in \Theta_{t-T:t}$, the corresponding natural parameters by ϕ_θ and, accordingly, let $\phi_{\Delta_{t-T:t}}$ and $\phi_{\Xi^j_{t-T:t}}$ be the natural parameters of global variables and local variables, respectively. Clearly, the maximization of (48) with respect to the PDF $q(\Theta_{t-T:t})$ can be equivalently re-expressed with respect to the corresponding natural parameters $\phi_{\Theta_{t-T:t}}$ partitioned into $\phi_{\Delta_{t-T:t}}$ and $\{\phi_{\Xi^j_{t-T:t}}\}$.

Then, as shown in [25], the overall maximization (48) can be recast into the following two coupled maximizations

$$\phi^*_{\Delta_{t-T:t}} = \arg\max_{\phi_{\Delta_{t-T:t}}} \sum_{j\in\mathcal{S}} \mathcal{L}_j\left(\phi_{\Delta_{t-T:t}}, \phi^*_{\Xi^j_{t-T:t}}\right) \tag{50}$$

$$\phi^*_{\Xi^j_{t-T:t}} = \arg\max_{\phi_{\Xi^j_{t-T:t}}} \mathcal{L}_j\left(\phi^*_{\Delta_{t-T:t}}, \phi_{\Xi^j_{t-T:t}}\right),\ j \in \mathcal{S} \tag{51}$$

to be solved with respect to the global $\phi_{\Delta_{t-T:t}}$ and, respectively, local $\phi_{\Xi^j_{t-T:t}}$ natural parameters. In a distributed setting, the maximization (50) cannot be directly solved, as no fusion center aggregates all local natural parameters. However, a distributed fixed-point method can be applied to iteratively solve (50)-(51) by exchanging information with neighbors [25], [28].

In our case, the resulting DVB approach consists of the following steps.

- Initialize the natural parameters $\phi^{*,0}_{\Xi^j_{t-T:t}}$ of the local variables and the local natural parameter $\phi^{j,0}_{\Delta_{t-T:t}}$ of the global variables at each node j.
- Perform a VB iteration with respect to the local natural parameters of the global variables

$$\phi^{j,*,n+1}_{\Delta_{t-T:t}} = \arg\max_{\phi^j_{\Delta_{t-T:t}}} \mathcal{L}_j\left(\phi^j_{\Delta_{t-T:t}}, \phi^{*,n}_{\Xi^j_{t-T:t}}\right),\ j \in \mathcal{S}. \tag{52}$$

- Perform, at each node j, a consensus step on the local natural parameters of the global variables

$$\begin{aligned}\varphi^{j,n+1}_{\Delta_{t-T:t}} &= \phi^{j,n}_{\Delta_{t-T:t}} + \alpha^{n+1}(\phi^{j,*,n+1}_{\Delta_{t-T:t}} - \phi^{j,n}_{\Delta_{t-T:t}})\\ \phi^{j,n+1}_{\Delta_{t-T:t}} &= \sum_{\ell\in\mathcal{S}_j} \varrho_{j,\ell}\,\varphi^{\ell,n+1}_{\Delta_{t-T:t}}\end{aligned} \tag{53}$$

  with step-size $\alpha^{n+1} > 0$ and consensus weights $\varrho_{j,\ell} > 0$ satisfying $\sum_{\ell\in\mathcal{S}_j}\varrho_{j,\ell} = 1$.
- Perform a VB iteration with respect to the natural parameters of the local variables for each j

$$\phi^{*,n+1}_{\Xi^j_{t-T:t}} = \arg\max_{\phi_{\Xi^j_{t-T:t}}} \mathcal{L}_j\left(\phi^{j,n+1}_{\Delta_{t-T:t}}, \phi_{\Xi^j_{t-T:t}}\right). \tag{54}$$

Then, the centralized R-VB-MHE-BG algorithm can be extended to a distributed setting by following the above DVB procedure, resulting into the following R-DVB-MHE-BG algorithm consisting of three steps. In particular, the maximizations in (52) and (54) can be solved in a similar way as in Section III as detailed in the following steps.

1) VB Iteration With Respect to the Local Natural Parameters of the Global Variables: aims to accomplish local estimation, in each node j, of the natural parameters of the global variables $x_{t-T:t}$ and Q according to (52). The difference with respect to the centralized case is that we use the local ELBO instead of the global ELBO. Hence, due to the exponentiation of the likelihood to the number J of agents in (49), measurements should be replicated J times. This step is carried out exploiting Propositions 1-2. In particular, state estimates and covariances can be obtained from the natural parameters of the state trajectory, namely $\omega^{j,n+1}_{t-T:t}$ and $\Omega^{j,n+1}_{t-T:t}$. Let us denote the local natural parameters as $\omega^{j,*,n+1}_{t-T:t}$ and $\Omega^{j,*,n+1}_{t-T:t}$. Then, it can be observed that $\omega^{j,*,n+1}_{t-T:t}$ and $\Omega^{j,*,n+1}_{t-T:t}$ are actually decomposed into innovation and, respectively, prior pair according to

$$\omega^{j,*,n+1}_{t-T:t} = \begin{bmatrix}\omega^{j,*,n+1}_{1,t-T+1:t}\\ 0\end{bmatrix} + \begin{bmatrix}0\\ \omega^{j,*,n+1}_{2,t-T}\end{bmatrix} \tag{56}$$

$$\Omega^{j,*,n+1}_{t-T:t} = \begin{bmatrix}\Omega^{j,*,n+1}_{1,t-T+1:t} & 0\\ 0 & 0\end{bmatrix} + \Omega^{j,*,n+1}_{2,t-T:t} \tag{57}$$

$$\Omega^{j,*,n+1}_{2,t-T:t} = \begin{bmatrix} \Phi^{j,*,n+1} & -\Phi^{j,*,n+1}F & \cdots & 0 & 0\\ -F'\Phi^{j,*,n+1} & \Phi^{j,*,n+1} + F'\Phi^{j,*,n+1}F & \ddots & & 0\\ \vdots & \ddots & \ddots & & \vdots\\ 0 & & & \Phi^{j,*,n+1} + F'\Phi^{j,*,n+1}F & -\Phi^{j,*,n+1}F\\ 0 & 0 & \cdots & -F'\Phi^{j,*,n+1} & F'\Phi^{j,*,n+1}F + \Omega^{j,*,n+1}_{2,t-T}\end{bmatrix} \tag{55}$$

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where $\omega_{2,t-T}^{j,*,n+1}$ is given by

$$\omega_{2,t-T}^{j,*,n+1} = \omega_{2,t-T}^{j,n} \tag{58}$$

while $\Omega_{2,t-T:t}^{j,*,n+1}$ is given in (55) with $\Phi^{j,*,n+1}$ and $\Omega_{2,t-T}^{j,*,n+1}$ defined as

$$\Phi^{j,*,n+1} = m_t^{j,n}(M_t^{j,n})^{-1} \tag{59}$$

$$\Omega_{2,t-T}^{j,*,n+1} = \Omega_{2,t-T}^{j,n}\,. \tag{60}$$

Conversely, due to exponentiation by J of the likelihood in (49) and the fact that such exponentiation of an exponential class PDF amounts to multiplication by J of its natural parameters, the innovation pair $\omega_{1,t-T+1:t}^{j,*,n+1}, \Omega_{1,t-T+1:t}^{j,*,n+1}$ is defined as

$$\omega_{1,t-T+1:t}^{j,*,n+1} = J \begin{bmatrix} (H^j)'\Psi_t^{j,n+1} y_t^j \\ (H^j)'\Psi_{t-1}^{j,n+1} y_{t-1}^j \\ \vdots \\ (H^j)'\Psi_{t-T+1}^{j,n+1} y_{t-T+1}^j \end{bmatrix} \tag{61}$$

$$\Omega_{1,t-T+1:t}^{j,*,n+1} = J \begin{bmatrix} (H^j)'\Psi_t^{j,n+1} H^j & 0 & \cdots & 0 \\ 0 & (H^j)'\Psi_{t-1}^{j,n+1} H^j & 0 & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & (H^j)'\Psi_{t-T+1}^{j,n+1} H^j \end{bmatrix} \tag{62}$$

where $\Psi_i^{j,n+1}$ can be computed by

$$\begin{aligned} \Psi_i^{j,n+1} = {} & (1 - \mathbb{E}^n[\eta_i^j])\, s_t^{j,n} (S_t^{j,n})^{-1} \\ & + \mathbb{E}^n[\eta_i^j]\, u_i^{j,n}\, (U_i^{j,n})^{-1}. \end{aligned} \tag{63}$$

In addition to the state trajectory $x_{t-T:t}$, the natural parameters of the PNCM Q (the other global variable) require to compute $m_t^{j,*,n+1}$ and $M_t^{j,*,n+1}$ as follows

$$\begin{aligned} m_t^{j,*,n+1} = {} & \bar{m}_{t-T}^j + T \\ M_t^{j,*,n+1} = {} & \bar{M}_{t-T}^j + \sum_{i=t-T+1}^{t} \Big[(\hat{x}_i^{j,n} - F\hat{x}_{i-1}^{j,n}) \\ & \times (\hat{x}_i^{j,n} - F\hat{x}_{i-1}^{j,n})' + P_i^{j,n} + F P_{i-1}^{j,n} F' \\ & - P_{i,i-1}^{j,n} F' - F P_{i-1,i}^{j,n} \Big]. \end{aligned} \tag{64}$$

2) Consensus on the Local Natural Parameters of the Global Variables: Given the local natural parameters of the global variables obtained through (56), (57) and (64), to get the global natural parameters, we follow the approach of [25] consisting of two steps.

First, obtain the local intermediate parameters $\varphi_{\omega_{1,t-T+1:t}}^{j,n+1}, \varphi_{\Omega_{1,t-T+1:t}}^{j,n+1}, \varphi_{\Phi}^{j,n+1}, \varphi_{m_t}^{j,n+1}, \varphi_{M_t}^{j,n+1}$ via the following gradient ascent step

$$\begin{aligned} \varphi_{\omega_{1,t-T+1:t}}^{j,n+1} &= \omega_{1,t-T+1:t}^{j,n} + \alpha^{n+1}(\omega_{1,t-T+1:t}^{j,*,n+1} - \omega_{1,t-T+1:t}^{j,n}) \\ \varphi_{\Omega_{1,t-T+1:t}}^{j,n+1} &= \Omega_{1,t-T+1:t}^{j,n} + \alpha^{n+1}(\Omega_{1,t-T+1:t}^{j,*,n+1} - \Omega_{1,t-T+1:t}^{j,n}) \\ \varphi_{\Phi}^{j,n+1} &= \Phi^{j,n} + \alpha^{n+1}(\Phi^{j,*,n+1} - \Phi^{j,n}) \\ \varphi_{m_t}^{j,n+1} &= m_t^{j,n} + \alpha^{n+1}(m_t^{j,*,n+1} - m_t^{j,n}) \\ \varphi_{M_t}^{j,n+1} &= M_t^{j,n} + \alpha^{n+1}(M_t^{j,*,n+1} - M_t^{j,n}). \end{aligned} \tag{65}$$

where the step size α^{n+1} is selected so as to provide a suitable trade-off between diffusion speed (first term) and learning speed (second term); to this end, the choice $\alpha^{n+1} = \frac{1}{d_0+\tau(n+1)}$, where $1 \le d_0$ and $0 < \tau < 1$, is suggested [25]. Secondly, to spread information through the network, carry out the following average consensus step

$$\begin{aligned} \omega_{1,t-T+1:t}^{j,n+1} &= \sum\nolimits_{\ell\in\mathcal{S}_j} \varrho_{j,\ell}\, \varphi_{\omega_{1,t-T+1:t}}^{\ell,n+1} \\ \Omega_{1,t-T+1:t}^{j,n+1} &= \sum\nolimits_{\ell\in\mathcal{S}_j} \varrho_{j,\ell}\, \varphi_{\Omega_{1,t-T+1:t}}^{\ell,n+1} \\ \Phi^{j,n+1} &= \sum\nolimits_{\ell\in\mathcal{S}_j} \varrho_{j,\ell}\, \varphi_{\Phi}^{\ell,n+1} \\ m_t^{j,n+1} &= \sum\nolimits_{\ell\in\mathcal{S}_j} \varrho_{j,\ell}\, \varphi_{m_t}^{\ell,n+1} \\ M_t^{j,n+1} &= \sum\nolimits_{\ell\in\mathcal{S}_j} \varrho_{j,\ell}\, \varphi_{M_t}^{\ell,n+1} \\ \omega_{2,t-T}^{j,n+1} &= \sum\nolimits_{\ell\in\mathcal{S}_j} \varrho_{j,\ell}\, \omega_{2,t-T}^{\ell,*,n+1} \\ \Omega_{2,t-T}^{j,n+1} &= \sum\nolimits_{\ell\in\mathcal{S}_j} \varrho_{j,\ell}\, \Omega_{2,t-T}^{\ell,*,n+1} \end{aligned} \tag{66}$$

Finally, the natural parameters of the state trajectory can be obtained as follows

$$\omega_{t-T:t}^{j,n+1} = \begin{bmatrix} \omega_{1,t-T+1:t}^{j,n+1} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_{2,t-T}^{j,n+1} \end{bmatrix} \tag{67}$$

$$\Omega_{t-T:t}^{j,n+1} = \begin{bmatrix} \Omega_{1,t-T+1:t}^{j,n+1} & 0 \\ 0 & 0 \end{bmatrix} + \Omega_{2,t-T:t}^{j,n+1} \tag{68}$$

where $\Omega_{2,t-T:t}^{j,n+1}$ can be formed as in (55) using the third and seventh equations in (66) instead of (59) and (60). From the above information form, we can get mean and covariance via

$$\hat{x}_{t-T:t}^{j,n+1} = (\Omega_{t-T:t}^{j,n+1})^{-1} \omega_{t-T:t}^{j,n+1} \tag{69}$$

$$P_{t-T:t}^{j,n+1} = (\Omega_{t-T:t}^{j,n+1})^{-1}. \tag{70}$$

3) VB Iteration With Respect to Natural Parameters of the Local Variables: so far, we have shown how to get the natural parameters $\phi_{\Delta_{t-T:t}}$ of the global variables by maximizing the global ELBO in (50). To update the natural parameters $\phi_{\Xi_{t-T:t}^j}$ of the local variables, we need to maximize the local ELBO in (51). The local VB updates follow Propositions 3-6 with e_t^j, f_t^j, ξ_i^j, U_i^j, u_i^j, S_t^j, s_t^j, and $\mathbb{E}[\cdot]$ replaced by $e_t^{j,n+1}$, $f_t^{j,n+1}$, $\xi_i^{j,n+1}$, $U_i^{j,n+1}$, $u_i^{j,n+1}$, $S_t^{j,n+1}$, $s_t^{j,n+1}$, and $\mathbb{E}^{n+1}[\cdot]$, respectively. Instead of D_i^j in (37), we use the following:

$$\begin{aligned} D_i^{j,n+1} = {} & (y_i^j - H^j \hat{x}_i^{j,n+1})(y_i^j - H^j \hat{x}_i^{j,n+1})' \\ & + H^j P_i^{j,n+1} (H^j)'. \end{aligned} \tag{71}$$

For the distributed algorithm, the calculation of the required expectations in the parameter updates is the same as in the centralized algorithm. A pseudocode of R-DVB-MHE-BG, summarizing the developments of this section, is provided in Algorithm 2.

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Algorithm 2: R-DVB-MHE-BG (window $t-T:t$, agent j)

Inputs: $\bar{x}^j_{t-T}$, $\bar{P}^j_{t-T}$, F, H^j, $y^j_{t-T+1:t}$, $\bar{M}^j_{t-T}$, $\bar{m}^j_{t-T}$, $\bar{S}^j_{t-T}$, $\bar{s}^j_{t-T}$, $\bar{U}^j_0$, $\bar{u}^j_0$, $\bar{e}^j_{t-T}$, $\bar{f}^j_{t-T}$, $\hat{Q}^j_{t-T}$, $\hat{R}^j_{t-T}$, N, T, ρ

VB local parameters of local variables initialization:

for each $j \in \mathcal{S}$ and $i = t-T+1:t$

$\mathbb{E}^0[\eta^j_i]$, $\mathbb{E}^0[\ln \pi^j]$, $\mathbb{E}^0[\ln(1-\pi^j)]$, $\mathbb{E}^0[(R^j)^{-1}]$, $\mathbb{E}^0[(\Sigma^j_i)^{-1}]$, $\mathbb{E}^0[\ln|R^j|]$, $\mathbb{E}^0[\ln|\Sigma^j_i|]$ are the same as Algorithm 1;

VB local parameters of global variables initialization:

Compute $\omega^{j,*,0}_{1,t-T+1:t}$ via (61);

Compute $\Omega^{j,*,0}_{1,t-T+1:t}$ via (62);

Set $\omega^{j,0}_{1,t-T+1:t} = \omega^{j,*,0}_{1,t-T+1:t}$ and $\Omega^{j,0}_{1,t-T+1:t} = \Omega^{j,*,0}_{1,t-T+1:t}$;

Set $\Phi^{j,0} = \bar{m}^j_{t-T}(\bar{M}^j_{t-T})^{-1}$;

Compute $\omega^{j,0}_{2,t-T} = (\bar{P}^j_{t-T})^{-1}\bar{x}^j_{t-T}$ and $\Omega^{j,0}_{2,t-T} = (\bar{P}^j_{t-T})^{-1}$;

Construct $\omega^{j,0}_{t-T:t}$ and $\Omega^{j,0}_{t-T:t}$ via (67)-(68) using $\omega^{j,0}_{1,t-T+1:t}$, $\Omega^{j,0}_{1,t-T+1:t}$, $\Phi^{j,0}$, $\omega^{j,*,0}_{2,t-T}$, $\Omega^{j,*,0}_{2,t-T}$;

Compute $\hat{x}^{j,0}_{t-T:t}$ and $P^{j,0}_{t-T:t}$ via (69)-(70) using $\Omega^{j,0}_{t-T:t}$ and $\omega^{j,0}_{t-T:t}$;

Compute $\bar{m}^{j,*,0}_t$ and $\bar{M}^{j,*,0}_{t-T}$ via (64);

Set $m^{j,0}_t = \bar{m}^{j,*,0}_t$ and $M^{j,0}_t = \bar{M}^{j,*,0}_{t-T}$;

VB moving horizon measurement update:

for $n = 0, 1, \ldots, N-1$ do

I. VB iterations with respect to global variables:

1: Compute $\omega^{j,*,n+1}_{t-T:t}$ and $\Omega^{j,*,n+1}_{t-T:t}$ via (56)-(63);

2: Compute $m^{j,*,n+1}_t$ and $M^{j,*,n+1}_t$ via (64);

II. Gradient ascent and consensus on local natural parameters of the global variables:

3: Compute $\varphi^{j,n+1}_{\omega_{1,t-T+1:t}}, \varphi^{j,n+1}_{\Omega_{1,t-T+1:t}}, \varphi^{j,n+1}_{\Phi}, \varphi^{j,n+1}_{m_t}, \varphi^{j,n+1}_{M_t}$ via (65);

Broadcast $\varphi^{j,n+1}_{\Delta_{t-T:t}}$

Receive $\varphi^{\ell,n+1}_{\Delta_{t-T:t}}$ from all $\ell \in \mathcal{S}_j \setminus \{j\}$

4: Compute $\omega^{j,n+1}_{1,t-T+1:t}, \Omega^{j,n+1}_{1,t-T+1:t}, \Phi^{j,n+1}_t, \omega^{j,n+1}_{2,t-T}, \Omega^{j,n+1}_{2,t-T}$, $m^{j,n+1}_t, M^{j,n+1}_t$ via (66);

5: Construct $\omega^{j,n+1}_{t-T:t}$ and $\Omega^{j,n+1}_{t-T:t}$ via (67)- (68);

6: Compute $\hat{x}^{j,n+1}_{t-T:t}$ and $P^{j,n+1}_{t-T:t}$ via (69)-(70);

III. VB iterations with respect to local variables:

7: Same as steps 8-17 in Algorithm 1 using $\hat{x}^{j,n+1}_{t-T:t}, P^{j,n+1}_{t-T:t}$;

end for

Moving horizon time-propagation:

Update $\bar{M}^j_{t-T+1}, \bar{m}^j_{t-T+1}, \bar{S}^j_{t-T+1}, \bar{s}^j_{t-T+1}, \bar{e}^j_{t-T+1}, \bar{f}^j_{t-T+1}$ using (45) for each j;

Compute $\bar{x}^j_{t-T+1}$, $\bar{P}^j_{t-T+1}$, and $\hat{Q}^j_{t-T+1}$ via (72);

Compute $\hat{R}^j_{t-T+1} = \frac{\bar{S}^j_{t-T+1}}{\bar{s}^j_{t-T+1} - n^j_y - 1}$;

Outputs: $\bar{x}^j_{t-T+1}$, $\bar{P}^j_{t-T+1}$, $\bar{M}^j_{t-T+1}$, $\bar{m}^j_{t-T+1}$, $\bar{S}^j_{t-T+1}$, $\bar{s}^j_{t-T+1}$, $\hat{Q}^j_{t-T+1}$, $\hat{R}^j_{t-T+1}$, $\bar{e}^j_{t-T+1}$, $\bar{f}^j_{t-T+1}$

B. Time-Propagation to the Next Window

In the distributed case, time-propagation of PNCM, INCM, and outlier probability takes the same form of (45), and the state time propagation from the current window to the next window is carried out as follows:

$$\begin{aligned}
\tilde{x}^j_{t-T+1} &= F\bar{x}^j_{t-T} \\
\hat{Q}^j_{t-T+1} &= \frac{M^j_{t-T+1}}{m^j_{t-T+1} - n_x - 1} \\
\tilde{P}^j_{t-T+1} &= F\bar{P}^j_{t-T}F' + \hat{Q}^j_{t-T+1} \\
\bar{P}^j_{t-T+1} &= ((\tilde{P}^j_{t-T+1})^{-1} + \Omega^{j,N}_{1,t-T+1})^{-1} \\
\bar{x}^j_{t-T+1} &= \bar{P}^j_{t-T+1}\left[(\tilde{P}^j_{t-T+1})^{-1}\tilde{x}^j_{t-T+1} + \omega^{j,N}_{1,t-T+1}\right].
\end{aligned} \tag{72}$$

where $\Omega^{j,N}_{1,t-T+1}$ and $\omega^{j,N}_{1,t-T+1}$ have the following form

$$\begin{aligned}
\Omega^{j,N}_{t-T+1} &= (H^j)'\Psi^{j,N}_{t-T+1}H^j \\
\omega^{j,N}_{t-T+1} &= (H^j)'\Psi^{j,N}_{t-T+1}y^j_{t-T+1}.
\end{aligned} \tag{73}$$

Remark 4: For the fusion strategy, R-DVB-MHE-BG differs from existing HCMCI-based algorithms [9]. While HCMCI relies solely on local measurements for variational Bayesian updates, R-DVB-MHE-BG progressively incorporates collected measurements for parameter estimation by employing gradient ascent and consensus fusion in each variational iteration. We highlight that in HCMCI-based methods, jointly estimating the PNCM and detecting outliers is challenging, as measurement anomalies can arise from either an increase in PNCM or from outliers. However, in R-DVB-MHE-BG, after several iterations, the algorithm can effectively distinguish between anomalies caused by PNCM and those due to outliers by leveraging multi-sensor information, thereby achieving higher estimation accuracy. However, this improvement may come with some additional computational effort. Specifically, the DVB-based fixed-point approach may require more variational consensus steps, resulting in a higher computational demand compared to HCMCI-based filters. Nonetheless, the computational requirements remain reasonable for practical applications.

V. Algorithm Based on Elliptical Distribution

The proposed MHE filter, based on the BG model, can actually be modified to easily accommodate various heavy-tailed elliptical noise distributions, such as the Student's t and multivariate Laplace, among others [15].

To this end, the likelihood (6) can be re-formulated as follows:

$$\begin{aligned}
&p\left(\{y^j_{t-T+1:t}\} \mid x_{t-T+1:t}, \{R^j\}, \{g(\lambda^j_i)\}\right) \\
&= \prod_{i=t-T+1}^{t}\prod_{j\in\mathcal{S}} \mathcal{N}(y^j_i; H^j x_i, R^j/g(\lambda^j_i))
\end{aligned} \tag{74}$$

where the scale factor variable $g(\lambda^j_i)$ can be defined in two ways. The first is $g(\lambda^j_i) = \lambda^j_i$, where the prior density can be: 1) $\mathcal{G}(\lambda^j_i; \frac{\nu}{2}, \frac{\nu}{2})$ for $\nu > 0$, 2) $\mathcal{G}(\lambda^j_i; \nu_1, \nu_2)$ for $\nu_1 > 0$ and $\nu_2 > 0$. The second is $g(\lambda^j_i) = (\lambda^j_i)^{-1}$, where the prior density can be: 3) $\mathcal{E}(\lambda^j_i; a)$ with $a = 1$ ($\mathcal{E}$ denotes exponential PDF), or 4) $\mathcal{N}^{-1}(\lambda^j_i; a, b)$ with $a > 0$ and $b > 0$ ($\mathcal{N}^{-1}$ denotes inverse Gaussian PDF). The above four options correspond respectively to Student's t (ST), Pearson-type VII (PVII), multivariate Laplace (ML), and normal inverse Gaussian (NIG) elliptical distribution [15].

The resulting filter should be initialized with the prior distribution $p\left(x_0, Q, \{R^j\}\right)$, similarly to (5). Using the prior distribution of the state trajectory $x_{t-T+1:t}$, as shown in (7), we then gather the measurements from each sensor j. Instead of (6), we can employ the likelihood function (74), selecting $g(\lambda^j_i)$ and $p(\lambda^j_i)$ according to Table I. By using different prior distributions $p(\lambda^j_i)$ in (74), the likelihood turns out to have a heavy-tailed character. This results into a joint PDF of the variable set $\Theta_{t-T:t} \triangleq \{x_{t-T:t}, Q, \{R^j\}, \{\lambda^j_i\}\}$ that can be

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TABLE I
SPECIFIC DEFINITIONS FOR ALGORITHMS BASED ON VARIOUS TYPES OF ELLIPTICAL DISTRIBUTION (ED) FOR MEASUREMENT NOISE MODELING

ED	$g(\lambda_i^j)$	Prior $p(\lambda_i^j)$	Posterior $q^*(\lambda_i^j)$	Posterior parameters
ST	λ_i^j	$\mathcal{G}\left(\lambda_i^j;\frac{\nu}{2},\frac{\nu}{2}\right)$	$\mathcal{G}\left(\lambda_i^j;\zeta_i^j,\iota_i^j\right)$	$\zeta_i^j=\frac{\nu}{2}+\frac{n_y^j}{2},\iota_i^j=\frac{\nu}{2}+\frac{1}{2}\operatorname{tr}\left(D_i^j\mathbb{E}[(R^j)^{-1}]\right)$
PVII	λ_i^j	$\mathcal{G}\left(\lambda_i^j;\nu_1,\nu_2\right)$	$\mathcal{G}\left(\lambda_i^j;\zeta_i^j,\iota_i^j\right)$	$\zeta_i^j=\nu_1+\frac{n_y^j}{2},\iota_i^j=\nu_2+\frac{1}{2}\operatorname{tr}\left(D_i^j\mathbb{E}[(R^j)^{-1}]\right)$
ML	$(\lambda_i^j)^{-1}$	$\mathcal{E}\left(\lambda_i^j;a\right)$	$\mathcal{GIG}\left(\lambda_i^j;\zeta_i^j,\iota_i^j,\delta_i^j\right)$	$\zeta_i^j=2a,\iota_i^j=\operatorname{tr}\left(D_i^j\mathbb{E}[(R^j)^{-1}]\right),\delta_i^j=1-\frac{n_y^j}{2}$
NIG	$(\lambda_i^j)^{-1}$	$\mathcal{N}^{-1}\left(\lambda_i^j;a,b\right)$	$\mathcal{GIG}\left(\lambda_i^j;\zeta_i^j,\iota_i^j,\delta_i^j\right)$	$\zeta_i^j=\frac{b}{a^2},\iota_i^j=\operatorname{tr}\left(D_i^j\mathbb{E}[(R^j)^{-1}]\right)+b,\delta_i^j=-\frac{n_y^j}{2}-\frac{1}{2}$

TABLE II
COMPUTATION OF THE POSTERIOR EXPECTATION FOR VARIOUS TYPES OF ELLIPTICAL DISTRIBUTION (ED) FOR MEASUREMENT NOISE MODELING

ED	$g(\lambda_i^j)$	$\mathbb{E}[g(\lambda_i^j)]$
ST, PVII	λ_i^j	ζ_i^j/ι_i^j
ML, NIG	$(\lambda_i^j)^{-1}$	$\frac{\sqrt{\zeta_i^j}\mathcal{K}_{\delta_i^j+1}\left(\sqrt{\zeta_i^j\iota_i^j}\right)}{\sqrt{\iota_i^j}\mathcal{K}_{\delta_i^j}\left(\sqrt{\zeta_i^j\iota_i^j}\right)}-\frac{2\delta_i^j}{\iota_i^j}$

used to find an approximate VB/DVB solution of the posterior $p(\Theta_{t-T:t}\mid\{y_{1:t}^j\})$. In the centralized case, the primary difference is that $\Psi_i^j=\mathbb{E}[g(\lambda_i^j)]\,\mathbb{E}[(R^j)^{-1}]=s_t^j(S_t^j)^{-1}\,\mathbb{E}[g(\lambda_i^j)]$ is used in place of Ψ_i^j defined in (22). Table I shows how, in the different cases, the priors are updated into the corresponding posteriors based on the current measurement y_i^j (recall that D_i^j is given by (37)). In addition, Table II provides details on the computation of the posterior expectation $\mathbb{E}[g(\lambda_i^j)]$ (see references [15], [17]). In the table, $\mathcal{K}_p(\cdot)$ represents the modified Bessel function of the second kind of order p. Consequently, the previous outlier detection procedure involving Propositions 3-5 is modified to exploit the posterior in $q^*(\lambda_i^j)$, which will be useful to scale the covariance R^j when an outlier is detected by the filter. For Proposition 6, (43) is replaced by

$$s_t^j=\bar{s}_{t-T}^j+T$$
$$S_t^j=\bar{S}_{t-T}^j+\sum_{i=t-T+1}^{t}\mathbb{E}[g(\lambda_i^j)]\,D_i^j\,. \tag{75}$$

On the other hand, in the distributed setting, the algorithm based on elliptic distributions uses

$$\begin{aligned}\Psi_i^{j,n+1}&=\mathbb{E}^{n+1}[g(\lambda_i^j)]\,\mathbb{E}^{n+1}[(R^j)^{-1}]\\&=s_t^{j,n+1}(S_t^{j,n+1})^{-1}\,\mathbb{E}^{n+1}[g(\lambda_i^j)]\end{aligned} \tag{76}$$

in (61)-(62), for each $i\in t-T+1:t$, instead of $\Psi_i^{j,n+1}$ given by (63).

Hereafter, the specific algorithm based on the ST elliptical distribution will be referred to as R-VB-MHE-ST (in contrast to R-VB-MHE-BG based on the BG model).

VI. SIMULATION RESULTS

Consider a 2-dimensional target tracking simulation case-study with target state vector $x=[p_x,p_y,v_x,v_y]'$, consisting of position and velocity components along x and y axes, in a multi-sensor setup with $F=\begin{bmatrix}I_2 & \Delta t I_2\\ 0 & I_2\end{bmatrix}$ and $H^j=\begin{bmatrix}I_2 & 0\end{bmatrix}$. Process noise w_t, inlier measurement noise n_t^j, and outlier measurement noise o_t^j are characterized by unknown PNCM $Q_t=q_t\begin{bmatrix}\frac{\Delta t^3}{3}I_2 & \frac{\Delta t^2}{2}I_2\\ \frac{\Delta t^2}{2}I_2 & \Delta t I_2\end{bmatrix}$, with $\Delta t=1$, $q_t=50(1+0.1\cos(4\pi t/K))m^2/s^3$, where K denotes the total simulation time. INCMs $R_t^j=(1+0.5\cos(4\pi t/K))R_0$, with $R_0=rI_2$ and $r=100m^2$, and ONCMs $\Sigma_t^j=\mu_t^jR_0$ with $\mu_t^j=100$ for $K/4\le t<K/2$, and $\mu_t^j=1000$ for $K/2\le t<3K/4$. For the ground-truth outlier probability, we set $\pi_t^j=1$ for $t<K/4$ and $t\ge 3K/4$, $\pi_t^j=0.1$ for $K/4\le t<K/2$, and $\pi_t^j=0.2$ for $K/2\le t<3K/4$. The reported simulation results are averaged over $N_{MC}=100$ independent Monte Carlo trials.

We consider the centralized Kalman Filter with true noise statistics (C-KF-TRUE) as a performance baseline and compare it with VB-MHE [8] ignoring the presence of outliers, R-VB-KF-GGSM [23], and the proposed R-VB-MHE-BG/ST. The parameters are set as follows: $\kappa=3$, $\bar{m}_0=\kappa+n_x+1$, $\bar{M}_0=\kappa\bar{Q}$ with $\bar{Q}=I_4$, $\bar{s}_0^j=\kappa+n_y^j+1$, $\bar{S}_0^j=\kappa\bar{R}$ with $\bar{R}=10R_0$, $\bar{u}_0^j=\kappa+n_y^j+1$, $\bar{U}_0^j=100\kappa\bar{R}$, $\bar{e}_0^j=1$, and $\bar{f}_0^j=1$. Additionally, we set the DOF parameter $\nu=5$ and the forgetting factor $\rho=0.99$. Notably, nominal $\bar{Q}$ and $\bar{R}$ differ significantly from the corresponding ground-truth values.

For the distributed filters, we consider D-VB-MHE [9] and R-D-VB-KF-GGSM [24] to be compared to our proposed R-DVB-MHE-BG/ST. We also consider D-KF-TRUE using the distributed KF [29] with true statistics as a performance baseline for distributed testing. All distributed filters use the same initial parameters as the centralized filters. For distributed filters, we set $d_0=1$, $\tau=0.1$, $N=10$, and $T=10$. The network consists of $J=10$ agents (sensors), with their communication links shown in Fig. 7 in the supplementary material. A total of 800 samples of simulation is carried out for each Monte Carlo trial. We evaluate the estimation accuracy using the root mean square error (RMSE) and time-averaged RMSE (ARMSE) for the state, and the square root of the normalized Frobenius norm (SRNFN) for covariance matrices [8]. From Fig. 2, it can be clearly seen that the current accuracy of VB-MHE in position and velocity significantly degrades under the influence of outliers. In contrast, the existing R-VB-KF-GGSM algorithm exhibits good position estimation accuracy but inadequate velocity estimation.

The proposed R-VB-MHE-BG/ST algorithms show superior estimation accuracy, with precision levels close to C-KF-TRUE.

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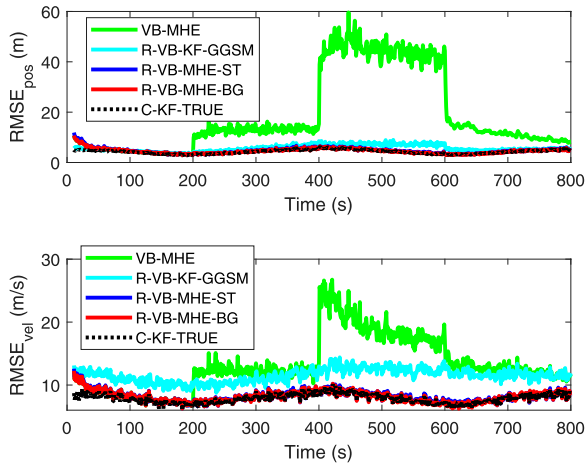


Fig. 2. RMSE$_{pos}$ and RMSE$_{vel}$ performance comparison.

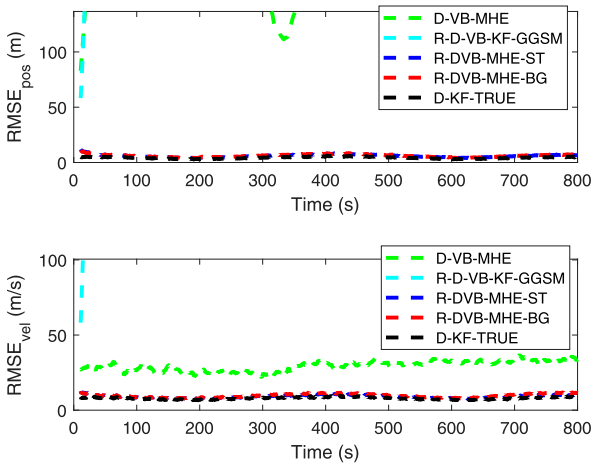


Fig. 4. RMSE$_{pos}$ and RMSE$_{vel}$ performance comparison.

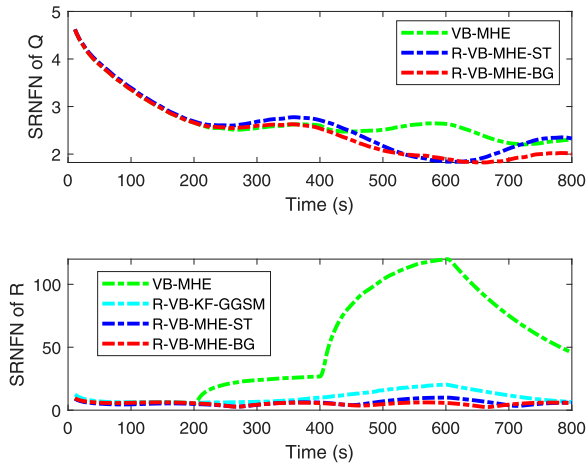


Fig. 3. SRNFN$_Q$ and SRNFN$_R$ performance comparison.

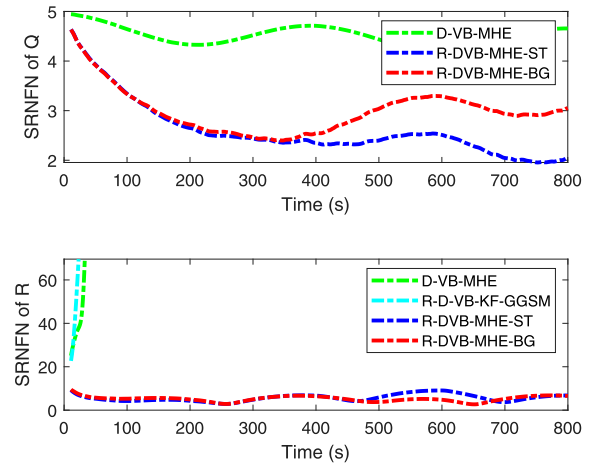


Fig. 5. SRNFN$_Q$ and SRNFN$_R$ performance comparison.

From Figs. 2 and 3, it can be seen that the accuracy degradation of VB-MHE is due to outliers affecting the estimation of covariance R_t, while other methods effectively estimate MNCM under outlier interference. R-VB-KF-GGSM achieves accurate position estimation by estimating PECM and using measurement-corrected positions. However, since R-VB-KF-GGSM relies on incorrect nominal PNCM and is able to estimate PECM only at each time step, due to the structural issues of the measurement equation, PECM lacks the velocity component of PNCM, thus resulting in significant velocity estimation errors. In contrast, the proposed method can estimate PNCM effectively. From Fig. 4, it can be seen that the accuracy of D-VB-MHE and R-D-VB-KF-GGSM is significantly deteriorated compared to the proposed R-DVB-MHE-BG/ST methods which yield almost the same performance as D-KF-TRUE. Fig. 5 shows that this performance deterioration is due to the influence of outliers on the estimation of covariance R_t in D-VB-MHE, thereby reducing its accuracy. R-D-VB-KF-GGSM only relies on local measurements to estimate PNCM and detect outliers and does not perform consensus on the PNCM parameters, which means that this method is inadequate to distinguish measurement changes caused by PNCM or outliers. When outliers occur, incorrect parameter estimates decrease the estimation accuracy of R-D-VB-KF-GGSM. Fortunately, the proposed methods can effectively estimate state and model parameters, thus getting better estimation accuracy. Fig. 6 plots the outlier probability π_t averaged over all sensors. The proposed filters can accurately estimate the time-varying outlier probability. No outliers are present during $t < K/4$ and $t > 3K/4$, resulting in an outlier probability close to zero. Between $K/4 \leq t < 3K/4$, outliers appear causing the outlier probability to fluctuate as the filter detects these outliers. Ultimately, the outlier probability converges to the true value. Table III reports the ARMSE of the considered centralized algorithms for different numbers N of VB iterations under fixed window size $T = 10$ showing that performance improves with increasing N, becoming near-optimal for $N \geq 2$. Table IV reports the ARMSE of the considered distributed algorithms. We note that since the DVB method requires a single data transmission per iteration, the algorithm performs the same

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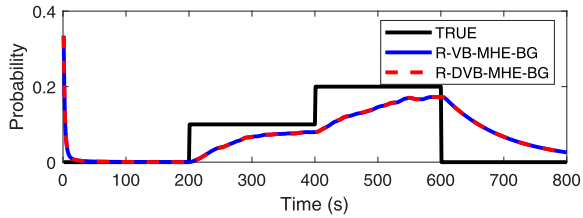


Fig. 6. Average outlier probability π_t over all sensors.

TABLE III
POSITION AND VELOCITY ARMSE OF THE CONSIDERED CENTRALIZED ALGORITHMS FOR DIFFERENT N

N	VB MHE		R-VB KF-GGSM		R-VB MHE-ST		R-VB MHE-BG	
	pos	vel	pos	vel	pos	vel	pos	vel
2	18.27	13.09	13.20	13.78	6.01	8.85	6.69	8.99
4	18.21	13.05	5.83	11.65	4.72	8.22	4.63	8.14
6	18.18	13.01	5.72	11.64	4.71	8.20	4.56	8.11
8	18.03	12.98	5.68	11.62	4.71	8.19	4.55	8.08
10	17.96	12.95	5.71	11.62	4.71	8.18	4.55	8.07
20	**18.46**	**12.91**	**5.59**	**11.58**	**4.71**	**8.15**	**4.55**	**8.05**

TABLE IV
POSITION AND VELOCITY ARMSE OF THE CONSIDERED DISTRIBUTED ALGORITHMS FOR DIFFERENT L AND N, WHERE $L = N$

N (L)	D-VB MHE		R-D-VB KF-GGSM		R-DVB MHE-ST		R-DVB MHE-BG	
	pos	vel	pos	vel	pos	vel	pos	vel
2	571.1	31.09	1012	43.50	26.81	202.9	29.53	410.1
4	354.5	29.26	592.1	39.59	12.73	63.12	14.62	185.3
6	245.1	29.29	987.1	44.89	7.68	18.14	8.94	29.89
8	241.7	29.28	1320	46.16	6.51	10.56	7.02	13.60
10	216.6	29.23	1022	46.72	6.15	9.60	6.46	10.77
20	**183.0**	**28.41**	**645**	**41.43**	**5.52**	**8.84**	**5.59**	**9.12**

TABLE V
POSITION AND VELOCITY ARMSE OF THE CONSIDERED CENTRALIZED ALGORITHMS FOR DIFFERENT T

T	R-VB-MHE-ST		R-VB-MHE-BG	
	pos	vel	pos	vel
1	100.16	23.70	137.27	24.10
2	19.71	13.40	13.19	11.95
3	6.38	9.25	5.72	8.87
4	5.13	8.53	4.87	8.26
5	4.88	8.36	4.67	8.21
10	**4.75**	**8.25**	**4.57**	**8.16**

TABLE VI
POSITION AND VELOCITY ARMSE OF THE CONSIDERED DISTRIBUTED ALGORITHMS FOR DIFFERENT T

T	R-DVB-MHE-ST		R-DVB-MHE-BG	
	pos	vel	pos	vel
1	97.94	23.53	125.42	23.55
2	17.83	12.94	12.28	11.69
3	6.69	9.35	6.39	9.34
4	5.83	8.93	5.82	9.12
5	5.61	8.76	5.61	8.96
10	**5.56**	**8.70**	**5.56**	**8.80**

TABLE VII
PRECISION AND RECALL OF THE CONSIDERED FILTERS

	Precision	Recall
R-VB-KF-GGSM	99.9405%	92.5706%
R-VB-MHE-ST	99.2606%	98.2363%
R-VB-MHE-BG	99.8953%	97.8597%
R-D-VB-KF-GGSM	–	28.8697%
R-DVB-MHE-ST	98.7713%	98.2327%
R-DVB-MHE-BG	99.9854%	96.2034%

number of variational iterations and consensus iterations, i.e. $N = L$. It can be seen that, compared to the centralized method, the proposed distributed method requires more iterations to converge. After three iterations, its performance is significantly better than the other algorithms and it basically converges after five iterations. Tables V and VI present the position and velocity ARMSEs provided by the centralized and distributed algorithms for different moving horizon sizes T and a fixed number of VB iterations ($N = 20$). The results show that the accuracy of both the proposed ST and BG algorithms improves with increasing T, becomimg near-optimal for $T \geq 4$.

Although we generate measurement noise data using Gaussian distributions with different (INCM and ONCM) covariances, due to the zero mean, it is still possible to generate normal values when ONCM is larger. Hence, in the outlier performance analysis, we use a χ^2-test with a 0.1% threshold to mark outliers. At each time, the squared normalized innovation error between each sensor measurement and true position

$$d_t^j = (y_t^j - H^j x_t)'(R_t^i)^{-1}(y_t^j - H^j x_t) \tag{77}$$

follows a χ^2 distribution [16] with threshold 18.421 corresponding to 0.1% outlier probability. When d_t^j exceeds the threshold, the current measurement is marked as an outlier.

Subsequently, precision and recall metrics [22], defined in terms of true positives (TP), false positives (FP), and false negatives (FN), are reported in Table VII. These metrics are computed by averaging over time, agents, and Monte Carlo trials, as follows:

$$Precision = \frac{1}{K\,J\,N_{MC}} \sum_{i=1}^{N_{MC}} \sum_{j=1}^{J} \sum_{t=1}^{K} \frac{TP_t^{j,i}}{TP_t^{j,i} + FP_t^{j,i}} \tag{78}$$

$$Recall = \frac{1}{K\,J\,N_{MC}} \sum_{i=1}^{N_{MC}} \sum_{j=1}^{J} \sum_{t=1}^{K} \frac{TP_t^{j,i}}{TP_t^{j,i} + FN_t^{j,i}}. \tag{79}$$

Precision measures the ability to detect outliers, with higher values indicating fewer false alarms. Recall reflects the comprehensiveness of detecting outliers, with higher values indicating fewer missed detections. The sum of $TP_t^{j,i}$ and $FN_t^{j,i}$ represents the total number of outliers in the ground truth at time t, agent j, and Monte Carlo trial i. As we can see from Table VII, the BG-based filter exhibits fewer false alarms but more missed detections compared to the ST-based filter, with a wider gap in the distributed case. These results highlight the differences between BG and ST-based filters which can be useful in selecting the most appropriate filter for practical applications.

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TABLE VIII
AVERAGE COMPUTING TIME PER RECURSION FOR DIFFERENT CENTRALIZED ALGORITHMS

Algorithms	Computing Time [ms]
C-KF-TRUE	0.099
R-VB-KF-GGSM	1.704
VB-MHE	10.856
R-VB-MHE-ST	12.304
R-VB-MHE-BG	28.951

TABLE IX
AVERAGE COMPUTING TIME PER AGENT PER RECURSION FOR DIFFERENT DISTRIBUTED ALGORITHMS

Algorithms	Computing Time [ms]
D-KF-TRUE	0.225
R-D-VB-KF-GGSM	0.479
D-VB-MHE	4.679
R-DVB-MHE-ST	4.964
R-DVB-MHE-BG	6.812

Moreover, Table VIII reports the average computing time per recursion (in milliseconds) for various centralized algorithms that aggregate data from all sensors. Additionally, Table IX provides the average computing time (in milliseconds) per agent per recursion for different distributed algorithms. As observed, compared to existing KF-based methods (C-KF-TRUE, R-VB-KF-GGSM, D-KF-TRUE, and R-D-VB-KF-GGSM), the proposed algorithms introduce an increased computational load due to the implementation of a moving horizon window. However, compared to the adaptive VB-KF-GGSM filters, this increase is proportional to the window size and remains compatible with real-time implementation. On the other hand, leveraging a sequence of measurements within the moving horizon framework allows for a more effective identification of outliers as well as covariance estimation, as demonstrated by the previous results. Furthermore, when compared to existing MHE-based methods (VB-MHE and D-VB-MHE), the slight increase of computational burden is well rewarded by the improved ability to counteract outliers. Finally, we observe that the ST-based approach is slightly less computationally demanding than the BG-based one due to the smaller number of parameters to estimate.

VII. CONCLUSION

The paper focused on a relevant challenging goal in state estimation, i.e. mitigating the deleterious effects of measurement outliers while simultaneously adapting to the unknown time-varying noise characteristics. To effectively cope with these two conflicting objectives, we introduced a novel variational Bayesian-moving horizon estimation (VB-MHE) framework that, for a linear Gaussian system with outliers and unknown time-varying noise covariances, allows to jointly estimate the state trajectory along with several noise parameters (process and measurement noise covariances, outlier probability) over a moving horizon of fixed length exploiting all measurements collected over the horizon. The proposed VB-MHE approach enables direct estimation of the process noise covariance and, unlike other outlier-robust filters, has the additional capability of inferring the outlier probability from the available data. Besides the derivation of a centralized robust adaptive filter, we also developed a distributed filter by leveraging distributed variational Bayesian (DVB) and consensus tools. We demonstrated, via simulation experiments, that both proposed, centralized and distributed, robust adaptive filters satisfactorily mitigate the effects of outliers and adapt to unknown time-varying noise covariances, exhibiting performance very close to the optimal one provided by a Kalman filter with perfect outlier detection and knowledge of noise statistics.

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